Supply Chain Management (SCM) System Workflow

System Architecture & Lifecycle Modules

- Procurement & Import Cycle (Phase 1)
    - Manages Purchase Requisitions generated from factories or departments.
    - Collects and evaluates vendor Quotations for pricing and quality.
    - Generates Purchase Orders (PurchaseOrder) and tracks items via POLineItem.
    - Handles international trade and Direct Import modules through Letter of Credit (LC) issuance via commercial banking integrations.
- Inbound Logistics & GRN (Phase 2)
    - Tracks inbound shipments via cargo and transport logistics.
    - Records Goods Received Notes (GRN) and itemized GRNLineItem entries upon warehouse arrival.
    - Maintains strict inventory control by holding newly received stock under an initial "Under Inspection" quarantine block.
- Quality Control (QC) Matrix (Phase 3)
    - Executes automated/manual quality audits managed by QCInspector profiles.
    - Validates product parameters using standardized QCChecklist matrices (e.g., density and tensile strength tests).
    - Automates conditional routing: PASS triggers inventory approval, while FAIL routes stock to vendor return workflows.
- Inventory & Stock Movement (Phase 4)
    - Automatically provisions backend inventory records upon successful QC verification.
    - Records StockMovement logs tracking inventory transfer from the "QC Hold Room" to the primary "Available Storage Warehouse".

- Sales & Customer Order Management (Phase 5)
    - Processes customer demands via CustomerOrder entries managed by sales personnel (SalesOfficer).
    - Manages itemized order bookings (OrderLineItem) and generates commercial invoices upon order confirmation.
- Outbound Logistics & Delivery (Phase 6)
    - Manages dispatch scheduling and DeliveryTrip planning.
    - Assigns delivery fleets (Vehicle) and personnel (Driver) to fulfill shipments.
    - Closes delivery lifecycles through automated daily reporting and status verification.
- Geographical Mapping Module
    - Integrates relational entities (Country, Division, District, PoliceStation) to ensure accurate geographical routing, address management, and delivery tracking.